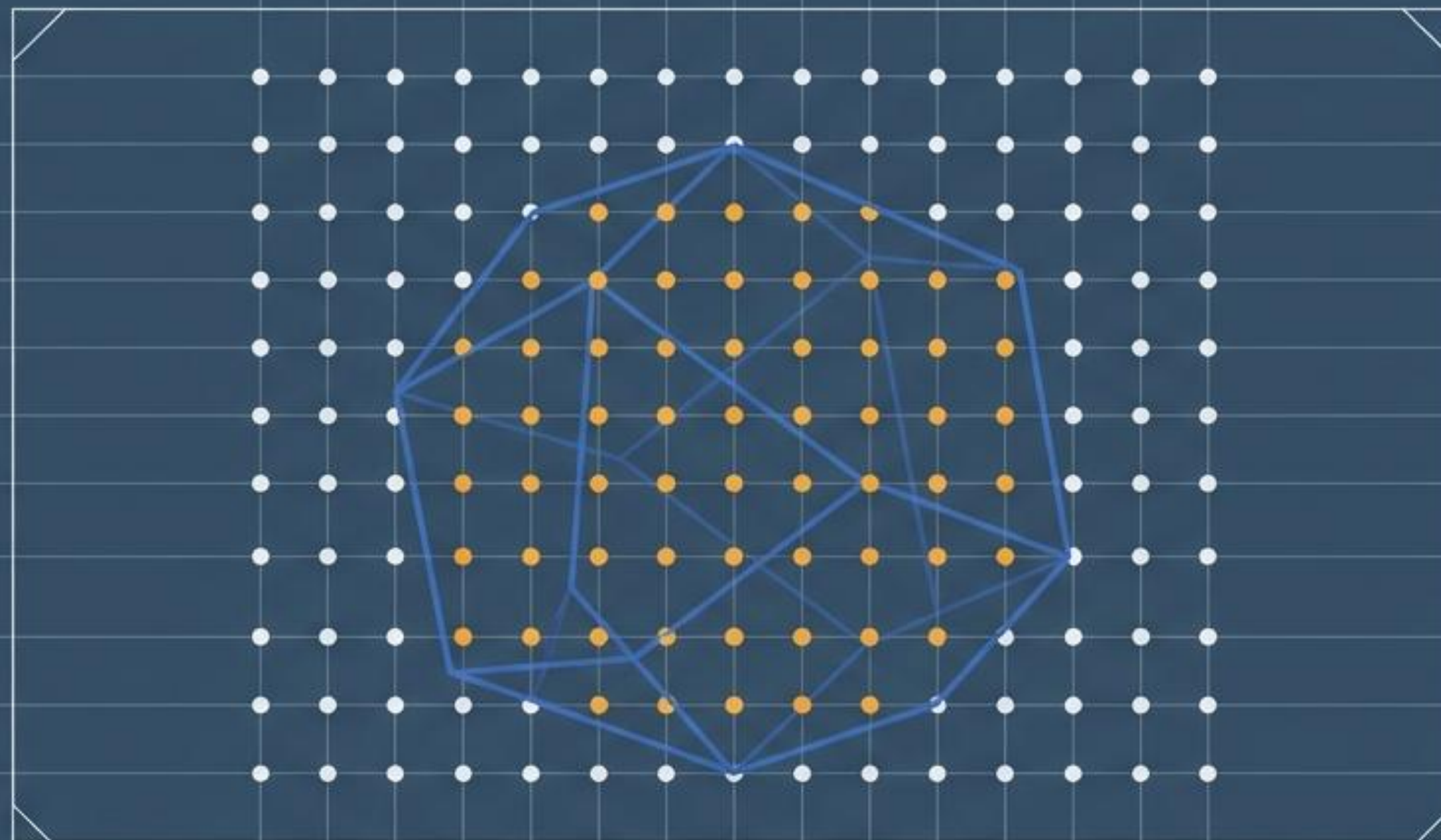




THE ARCHITECTURE OF OPTIMIZATION

FORMULATING LINEAR, INTEGER, AND MIXED-INTEGER PROGRAMMING MODELS

ACADEMIC PRESENTATION | MATHEMATICAL OPTIMIZATION MODELING

Presentation Outline

1. Linear Programming Basics

Establishing continuous variables and convexity.

2. Integer Variables & Logical Constraints

Introducing discrete constraints and binary logic.

3. Big-M & Linearization Techniques

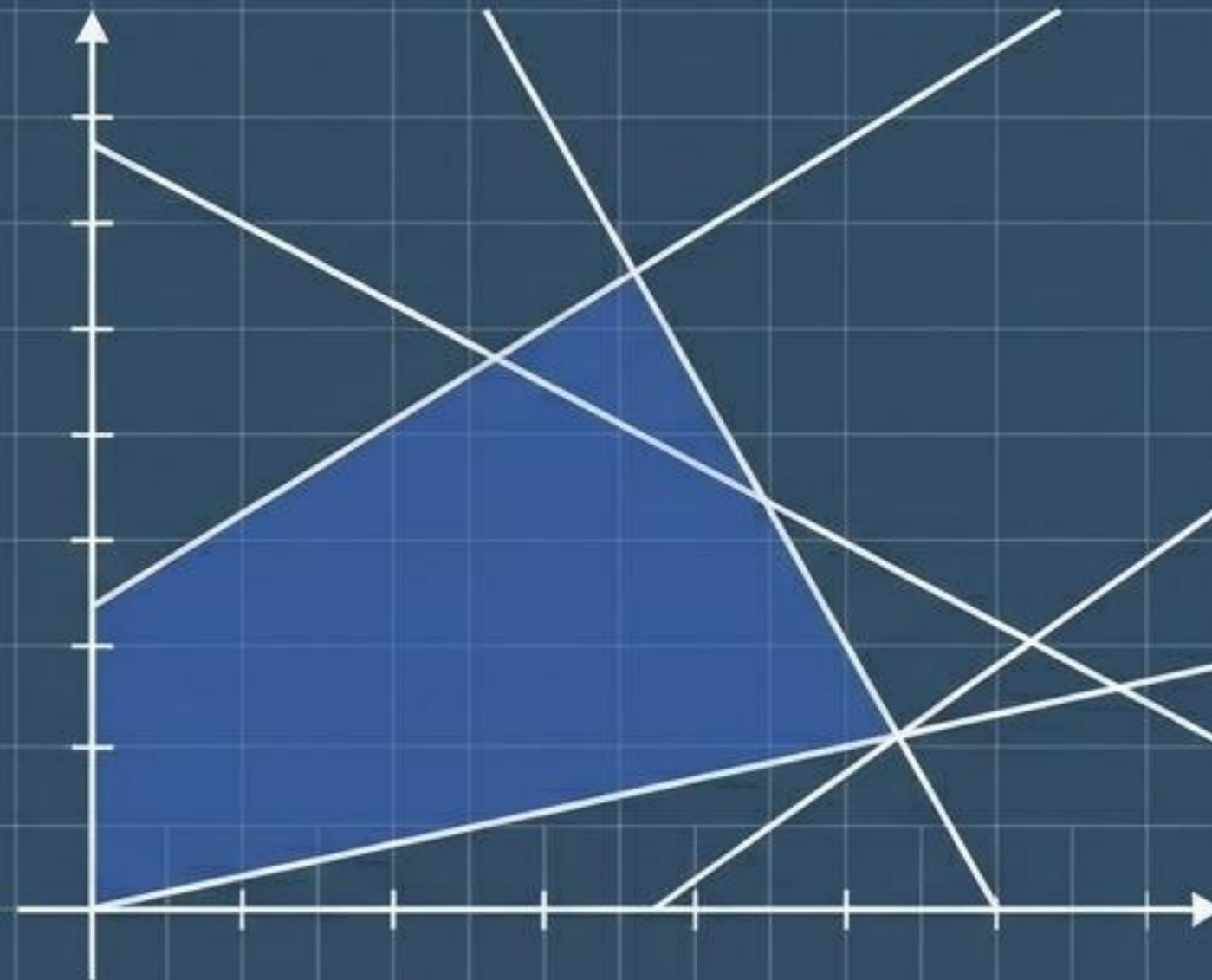
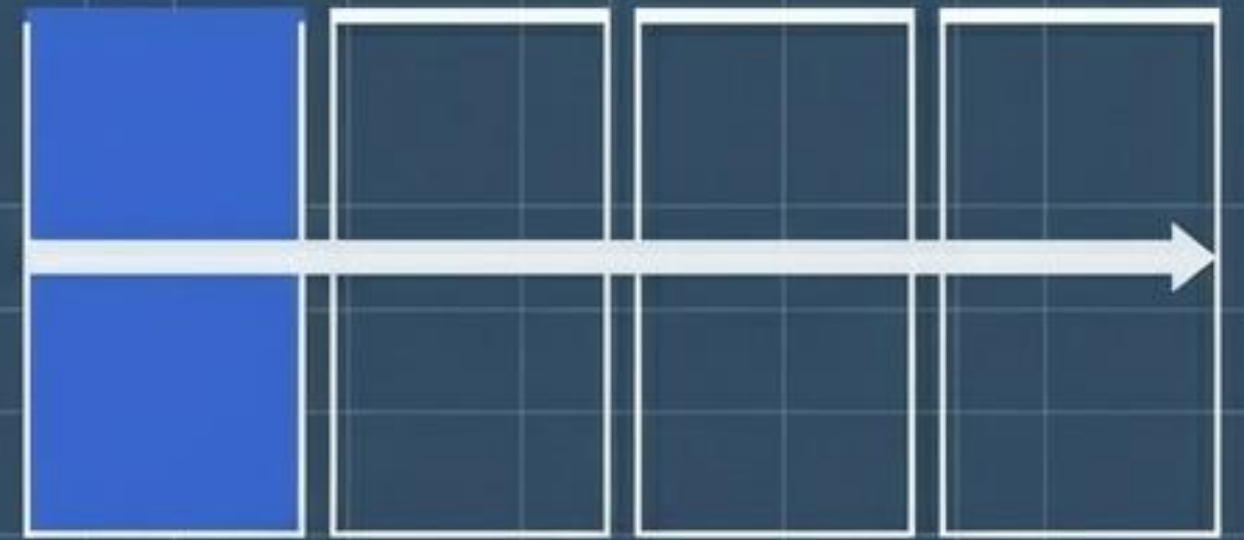
Bridging logical conditions into linear solvers.

4. Comprehensive MILP Example

Integrating all techniques into a unified model.

Act I: The Continuous Foundation

Before tackling complex discrete logic, we must establish the foundational framework of continuous linear optimization—where convexity guarantees global optimality.



Linear Programming (LP) Fundamentals

Decision Variables

Coordinates of the design vector:

$$\mathbf{x} \in \mathbb{R}^n$$

Must be continuous quantities (e.g., production volumes, time).

Objective Function

Performance criterion to minimize/maximize:

$$\min/\max \mathbf{c}^T \mathbf{x}$$

Must be strictly linear to ensure convexity.

Constraints

Resource limits or physical requirements:

$$\mathbf{Ax} \leq \mathbf{b}, \mathbf{Ax} = \mathbf{b}, \mathbf{Ax} \geq \mathbf{b}$$

Forms a convex polyhedron (the feasible region).

The Convex Guarantee

Because the feasible region is a convex polyhedron, any local optimum found by Simplex or Interior-Point algorithms is mathematically guaranteed to be a global optimum. Auxiliary variables are introduced to transform nonlinear expressions into this linear framework.

Example: LP Factory Production

[Problem]

A factory produces three products (P1, P2, P3). We must maximize profit under strict limits: 150 labor hours and 200 material units.

[Components]

Decision Variables:

$$x_1, x_2, x_3 \geq 0$$

(Continuous production quantities)

Profit/Unit: \$40 (P1), \$30 (P2), \$50 (P3)

Labor/Unit: 2 (P1), 1 (P2), 3 (P3)

Material/Unit: 3 (P1), 2 (P2), 4 (P3)

[Formulation]

Objective:

$$\max z = 40x_1 + 30x_2 + 50x_3$$

Subject To (s.t.):

$$2x_1 + 1x_2 + 3x_3 \leq 150$$

(Labor Limit Constraint)

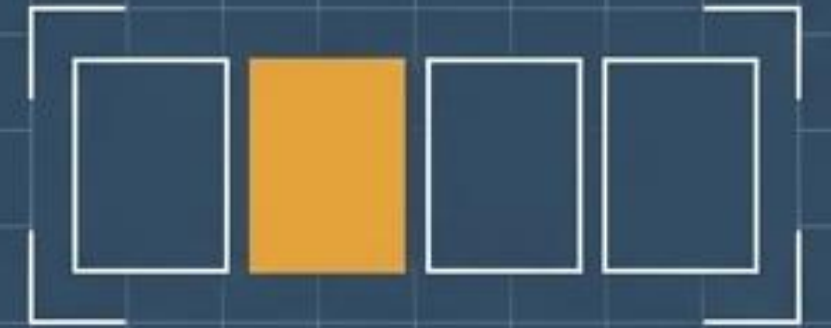
$$3x_1 + 2x_2 + 4x_3 \leq 200$$

(Material Limit Constraint)

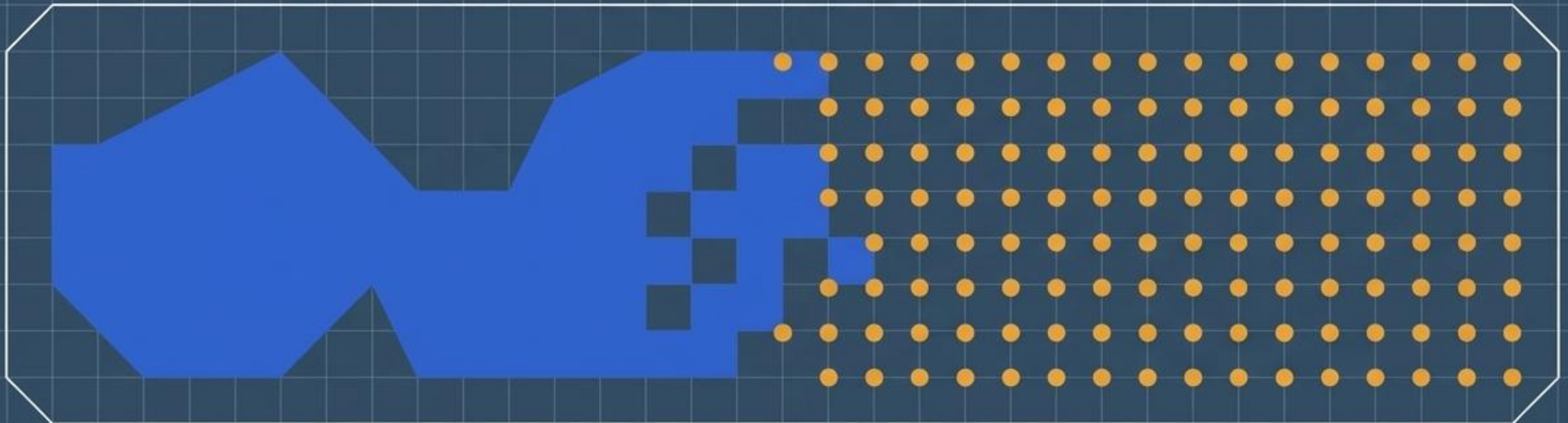
$$x_1, x_2, x_3 \geq 0$$

(Non-negativity)

Act II: The Discrete Complication



Continuous variables cannot represent binary choices like “yes/no” or “open/close”. We must introduce discrete integer models to capture real-world logical conditions, despite the computational cost.



Concept: Integer and Binary Programming

	Linear Programming (LP)	Integer Programming (IP)
Variable Domains	Continuous $\mathbb{R}^n$	Discrete $\mathbb{Z}$ or Binary $\{0, 1\}$
Feasible Region	Continuous convex polyhedron	Discrete, fragmented set of points
Complexity	Solvable in polynomial time	Combinatorial and generally NP-hard
Solvers	Simplex / Interior-Point JetBrains Mono	Branch-and-Bound / Cutting Planes JetBrains Mono

The Knapsack Problem

Binary variables ($x \in \{0, 1\}$) encode indivisible decisions—whether to select or reject specific projects or items under a rigid budget constraint. They form the foundation of logical modeling.

Translating Logic to Linear Inequalities

	Logical Concept	Boolean Logic	Linear Constraint
Category 1: Mutually Exclusive / Inclusion			
1	At least one is true	$x_1 \vee x_2 = 1$	$x_1 + x_2 \geq 1$
2	Exactly one is true	$(x_1=1, x_2=0) \vee (x_1=0, x_2=1)$	$x_1 + x_2 = 1$
3	Both must be true	$x_1 \wedge x_2 = 1$	$x_1 = 1, x_2 = 1$
Category 2: Implication (If-Then)			
4	If $x_1=0$, then $x_2=1$	$x_1=0 \Rightarrow x_2=1$	$x_2 \geq 1 - x_1$
5	If $x_1=0$, then at least two of (x_2, x_3) are 1	$x_1=0 \Rightarrow x_2+x_3 \geq 2$	$x_2 + x_3 \geq 2(1 - x_1)$
Category 3: Cardinality Bounds			
6	At most 'r' variables are 1	-	$\sum x_i \leq r$
7	If $x_1=0$, at least 'r' variables are 1	-	$\sum x_i \geq r(1 - x_1)$

Example: Store Opening with Logical Constraints

[Problem]

Evaluate 6 locations (L1-L6). Open a maximum of 4 stores. Minimize fixed and staffing costs.

[Logical Translation]

Rule: Max 4 stores $\rightarrow \sum y_i \leq 4$

Rule: L1 and L2 cannot both open $\rightarrow y_1 + y_2 \leq 1$

Rule: L3 and L4 must open together $\rightarrow y_3 - y_4 = 0$

[Components]

Binary Variables (Amber):

$y_i \in \{0,1\}$ (1 if Store i opens, 0 otherwise).

Continuous Variables (Cobalt):

$s_i \in \mathbb{Z}_{\geq 0}$ (Number of staff at Store i).

[Formulation]

Objective: $\min z = \sum (F_i y_i + c_i s_i)$

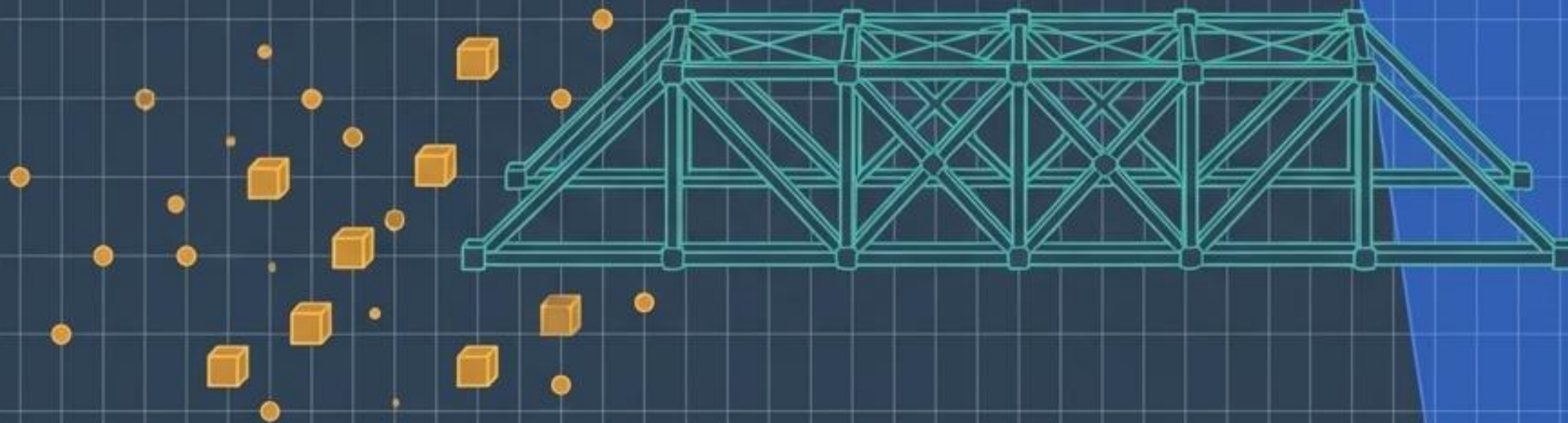
Staffing Bounds: $3y_i \leq s_i \leq 10y_i$

(Staff must be between 3 and 10 only if store is open).

Act III: The Bridge (MILP)

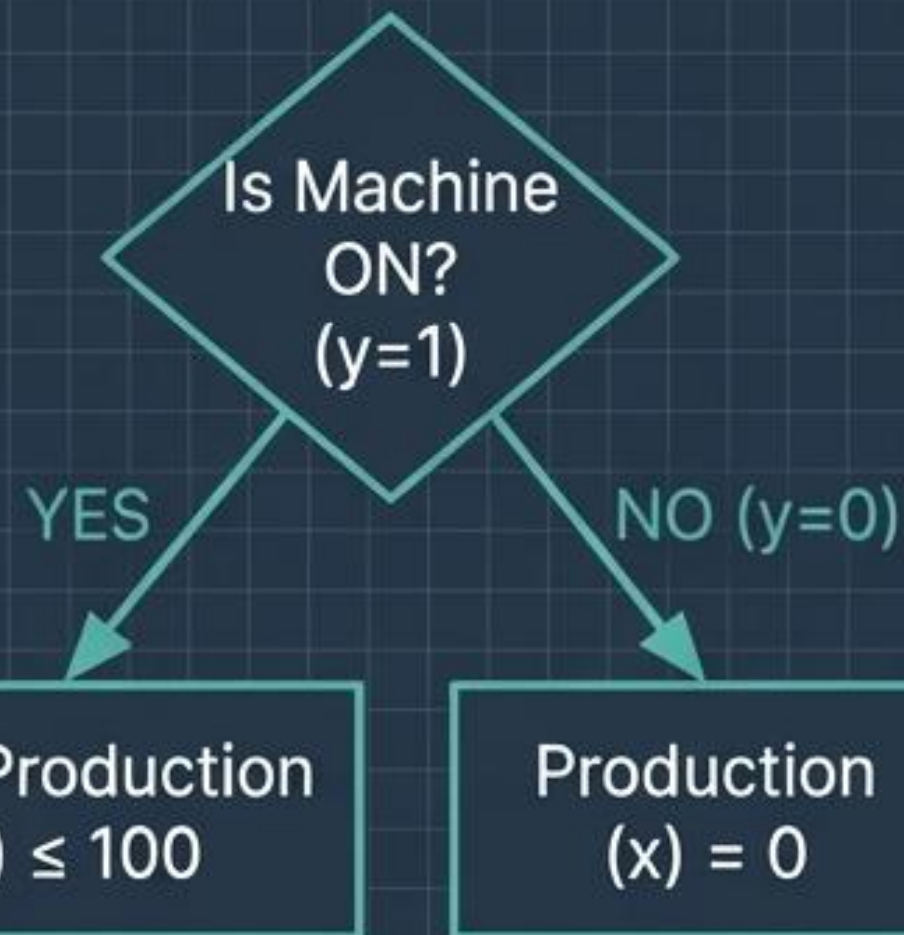


To model conditional operations and nonlinear costs within a rigid integer framework, we require specialized mathematical bridging techniques.



The Big-M Method: Activating Constraints Conditionally

Machine has 100 max capacity, 20 min capacity, 100 fixed cost, 5/unit variable cost.



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if y == 1 then
    20 <= x <= 100
else
    x == 0
```

Mathematical Formulation:

$$\begin{aligned} x &\leq 100y \\ x &\geq 20y \\ \min z &= 100y + 5x \end{aligned}$$

Note: The large constant M (here, 100) forces $x=0$ when $y=0$, and relaxes the constraint when $y=1$.

Linearizing Min/Max & Piecewise Functions

Linearizing MIN/MAX

Context: MILP solvers strictly prohibit $\min()$ or $\max()$ operators.

Formulation: To model $z = \min(x_A, x_B)$ natively:

Upper bounds:

$$z \leq x_A$$

$$z \leq x_B$$

Exact extraction (using Big-M and binary y):

$$z \geq x_A - M(1-y)$$

$$z \geq x_B - My$$

Piecewise Functions (Project Bonus)

Context: Project A grants a 10-unit bonus only if budget x_A exceeds 50.

Formulation: Introduce binary indicator $y_A \in \{0,1\}$ for $x_A > 50$.

$$x_A \leq 50 + M_A y_A$$

$$x_A \geq 50 y_A$$

$$\text{bonus}_A = 10 y_A$$

Act IV: The Capstone

Having mastered continuous constraints, binary logic, and Big-M techniques, we now integrate them into a unified Mixed-Integer Linear Programming capstone model.



Comprehensive Example: MILP Formulation

[Problem]

3 products (P1, P2, P3), 2 machines (M1, M2).
Products require exclusive machine assignment.
Machines have conditional setup times.
Maximize net profit.

[Formulation: Constraints]

Unique Assignment: $\sum_j a_{ij} = 1$

Conditional Logic: $a_{ij} \leq y_j$ (Assignment requires machine ON).

Big-M Capacity: $q_{\min_i} a_{ij} \leq x_{ij} \leq q_{\max_i} a_{ij}$ (Production bounded by assignment).

Total Labor: $10(y_1 + y_2) + \sum_i \sum_j \ell_i x_{ij} \leq 200$ (Setup + Variable).

[Components] (Variables)

$y_j \in \{0,1\}$: Machine j ON/OFF state (Amber).

$a_{ij} \in \{0,1\}$: Product i assigned to Machine j (Amber).

$x_{ij} \geq 0$: Units of i produced on j (Cobalt).

[Formulation: Objective]

$$\max z = \sum_i \sum_j p_i x_{ij} - \sum_j F_j y_j$$

(Net Profit = Revenue - Setup Costs).

The Architecture Realized

Pillar 1 The Continuous Foundation (LP)

Linear formulations are not merely restrictions; they are mathematical guarantees. Convexity ensures that local discoveries are global truths.

Pillar 2 The Discrete Complication (IP)

Binary variables provide the descriptive power to encode the nonlinear reality of human decisions—turning continuous flows into rigid, logical choices.

Pillar 3 The Synthesizing Bridge (MILP)

Techniques like Big-M are the structural engineering of optimization, forcing **conditional reality into linear solvers to solve complex, integrated systems.**

Mastery of mathematical optimization is the ability to architect seamless bridges between the continuous laws of physics and the discrete logic of business.